TEAM NAME:	Kindred
DOMAIN	Non-profit and NGO Operations
PROBLEM STATEMENT	Volunteer and Resource Coordination

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1. Preface:

- This document defines the requirements for the Volunteer and Resource Coordination System.
- Intended for developers, testers, admins, and end users.
- It explains what the system should do and how it should behave.

2.Introduction:

2.1 Need for the System:

NGOs face problems like:

Poor volunteer management

Lack of donation transparency

Difficulty in tracking resources

Manual and inefficient processes

2.2 Objective:

To develop a centralized platform that connects Volunteers, Donors, Beneficiaries, NGO administrators

3. Glossary

- **Volunteer:** Person providing service
- **Donor:** Person donating money/resources
- **Beneficiary:** Person receiving help
- **Administrator:** System controller
- **Resource:** Funds or materials donated

4. User Requirements

The system should allow users to:

- Access the platform through a web browser
- Register and login securely
- View information based on their role
- Receive notifications and updates
- Use the system easily without technical knowledge

5. System Requirements:

Functional requirements:

Functional requirements describe what a system is supposed to do, including the services it provides and the actions it performs in response to user input.

A. User Management:

- Volunteers can **register, login, and update profiles**
- NGOs/Admins can **verify volunteers**
- Different roles: **Volunteer, NGO/Administrator, Government Authority**

B. Volunteer Management:

- Register volunteer skills (medical, logistics, rescue, food supply)
- Assign volunteers to tasks based on **location & skills**
- Track volunteer availability (active/inactive)

C. Resource Management:

- Add resources (food, medicine, shelters, vehicles)
- Track resource quantity and location
- Allocate resources to affected areas

D. Task & Activity Coordination:

- Create tasks (rescue, distribution, medical help)
- Assign tasks to volunteers
- Update task status (pending, in-progress, completed)

E. Location & Mapping:

- Identify disaster-affected locations
- Show nearby volunteers and resources
- Route planning for delivery

F. Communication & Alerts:

- Send notifications to volunteers
- Emergency alerts to all users
- Messaging between volunteers and coordinators

G. Reporting & Monitoring:

- Generate reports (volunteers deployed, resources used)
- View real-time dashboards
- Audit logs for actions

Non-Functional Requirements:

Non- Functional requirements specify quality attributes and constraints of a system, such as performance, security and reliability.

A. Product Requirements:

1. Efficiency Requirements:

- System should handle **large number of users simultaneously**
- Resource allocation should be fast

2. Performance Requirements:

- Page response time ≤ **2 seconds**
- Notifications delivered in **real time**

3. Space Requirements:

- Efficient database storage for volunteer & resource data
- Optimize storage for images/documents

4. Usability Requirements:

- Simple UI for non-technical users
- Mobile-friendly interface
- Supports multiple languages

5. Dependability Requirements:

- System should be available **24/7**
- Data backup and recovery
- Minimal downtime during disasters

6. Security Requirements:

- Secure login and authentication
- Role-based access control
- Encrypt sensitive data (personal info, locations)

B. Organizational Requirements:

1. Operational Requirements:

- System should work on web and mobile
- Must run on cloud infrastructure

2. Environmental Requirements:

- Operates under low bandwidth conditions
- Supports use during power/network instability

3. Development Requirements:

- Developed using standard frameworks
- Easy to maintain and upgrade
- Modular design

C. External Requirements:

1. Regulatory Requirements:

- Follow government disaster-management guidelines
- Data protection compliance

2. Ethical Requirements:

- No misuse of volunteer personal data
- Fair task assignment without bias.

3. Legislative Requirements:

- Comply with IT laws and data privacy acts
- Maintain legal audit logs

4. Accounting Requirements:

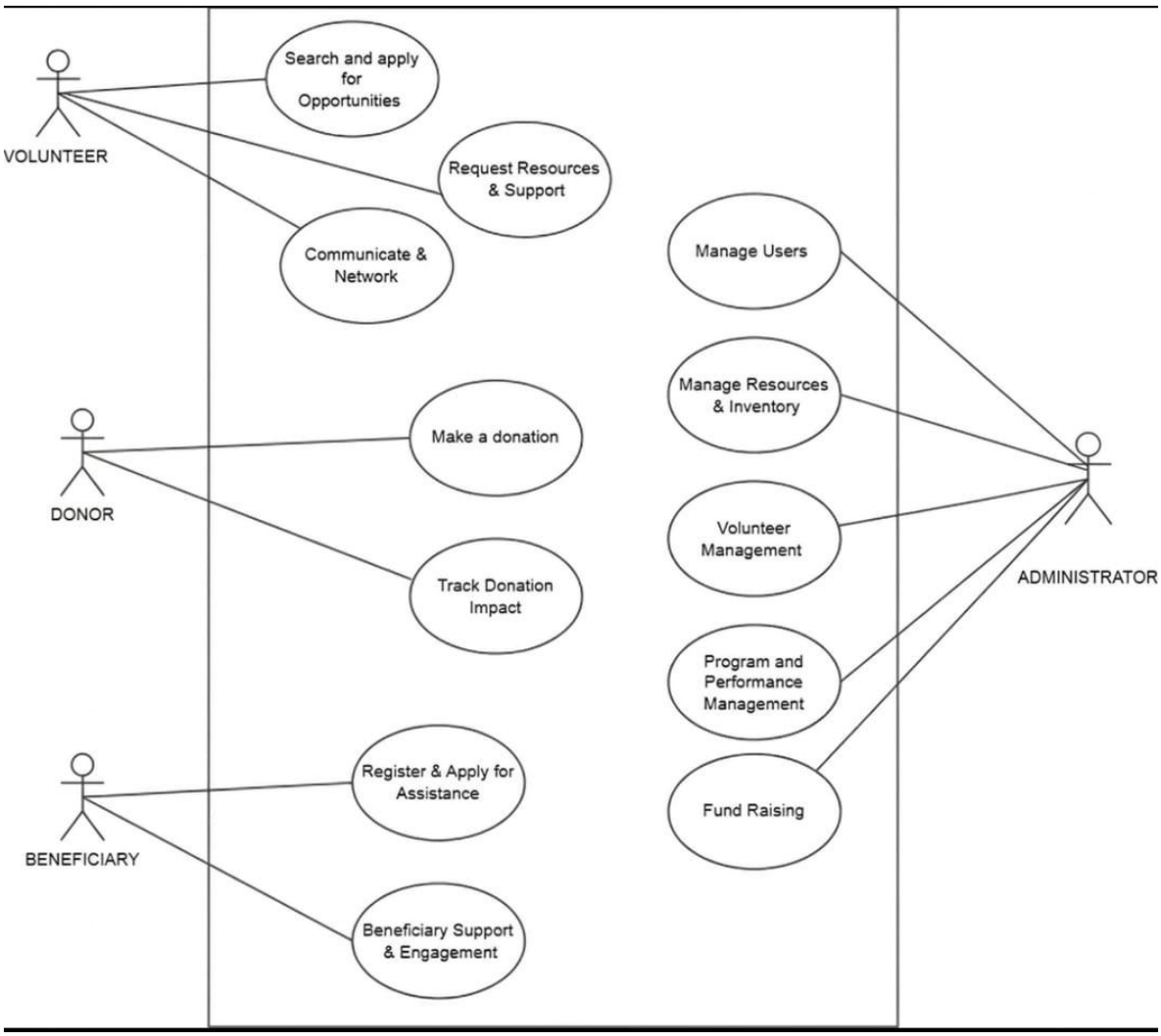
- Track donations and resource funding
- Generate financial usage reports

5. Safety / Security Requirements:

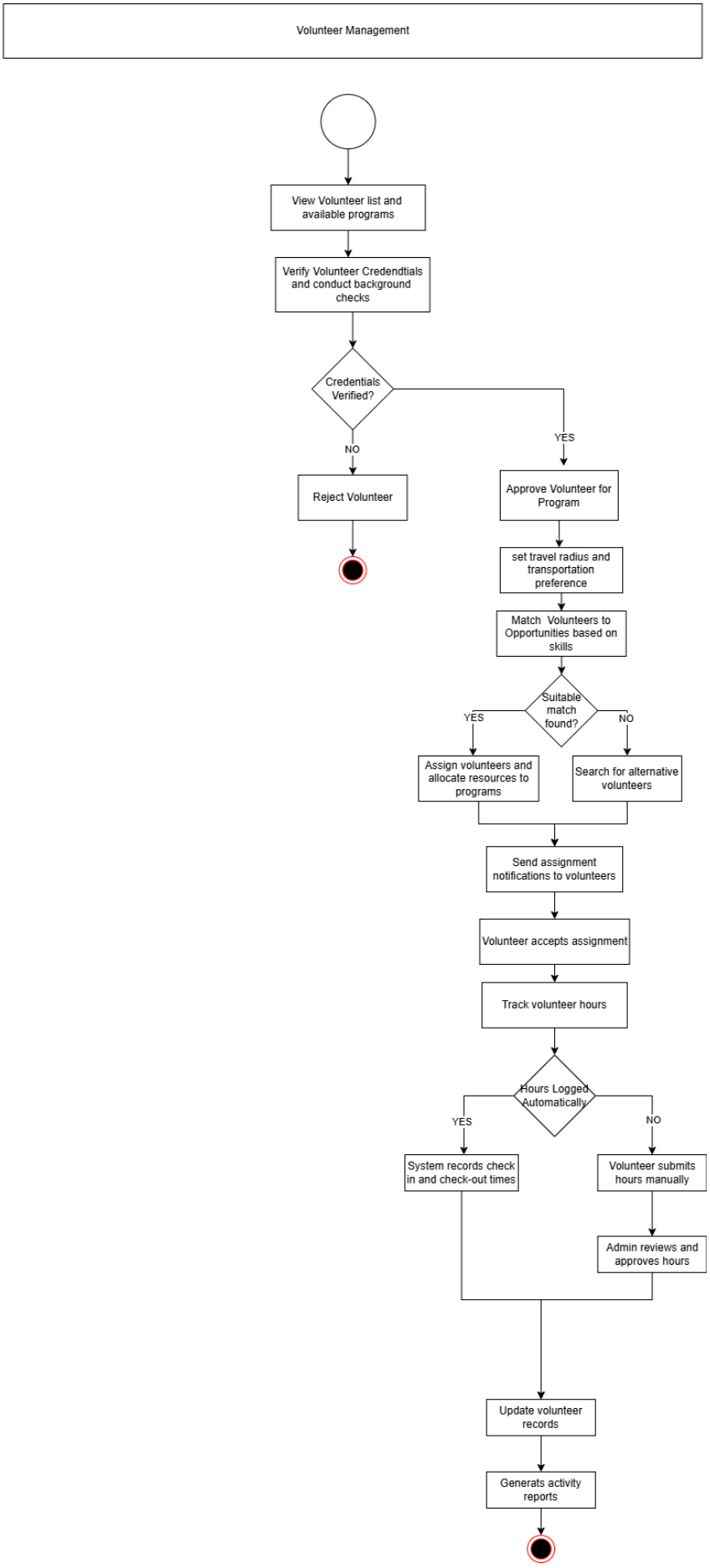
- Protect system from cyber attacks
- Prevent false emergency requests

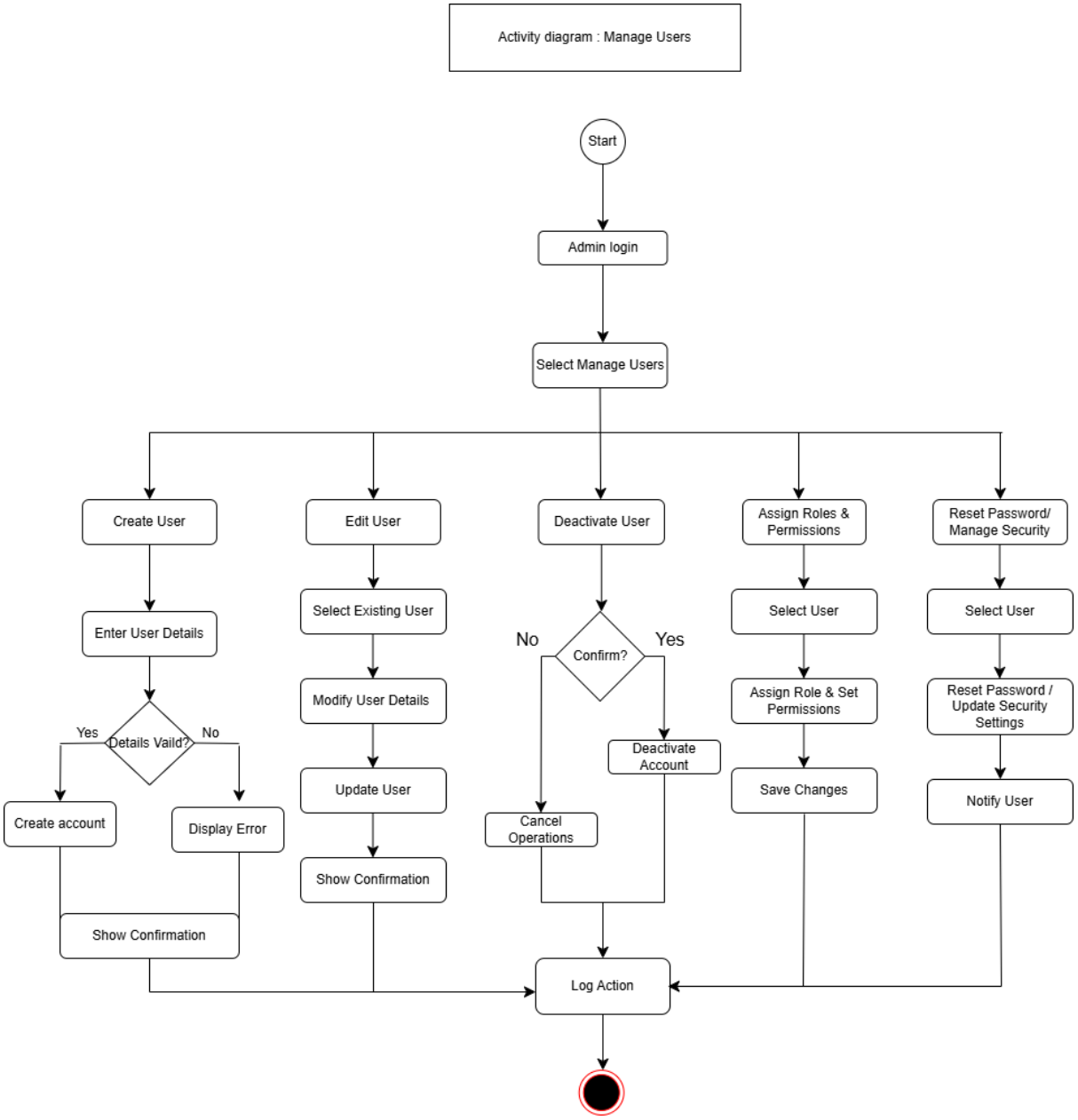
6.System Models

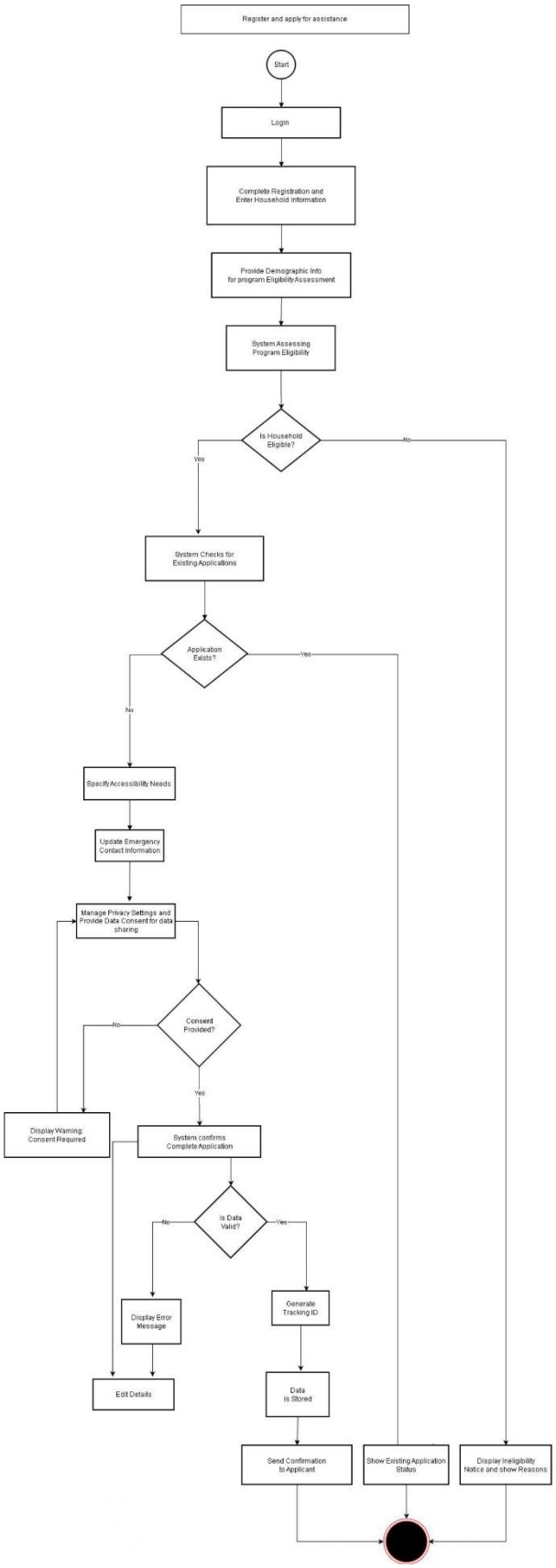
6.1 USE CASE:



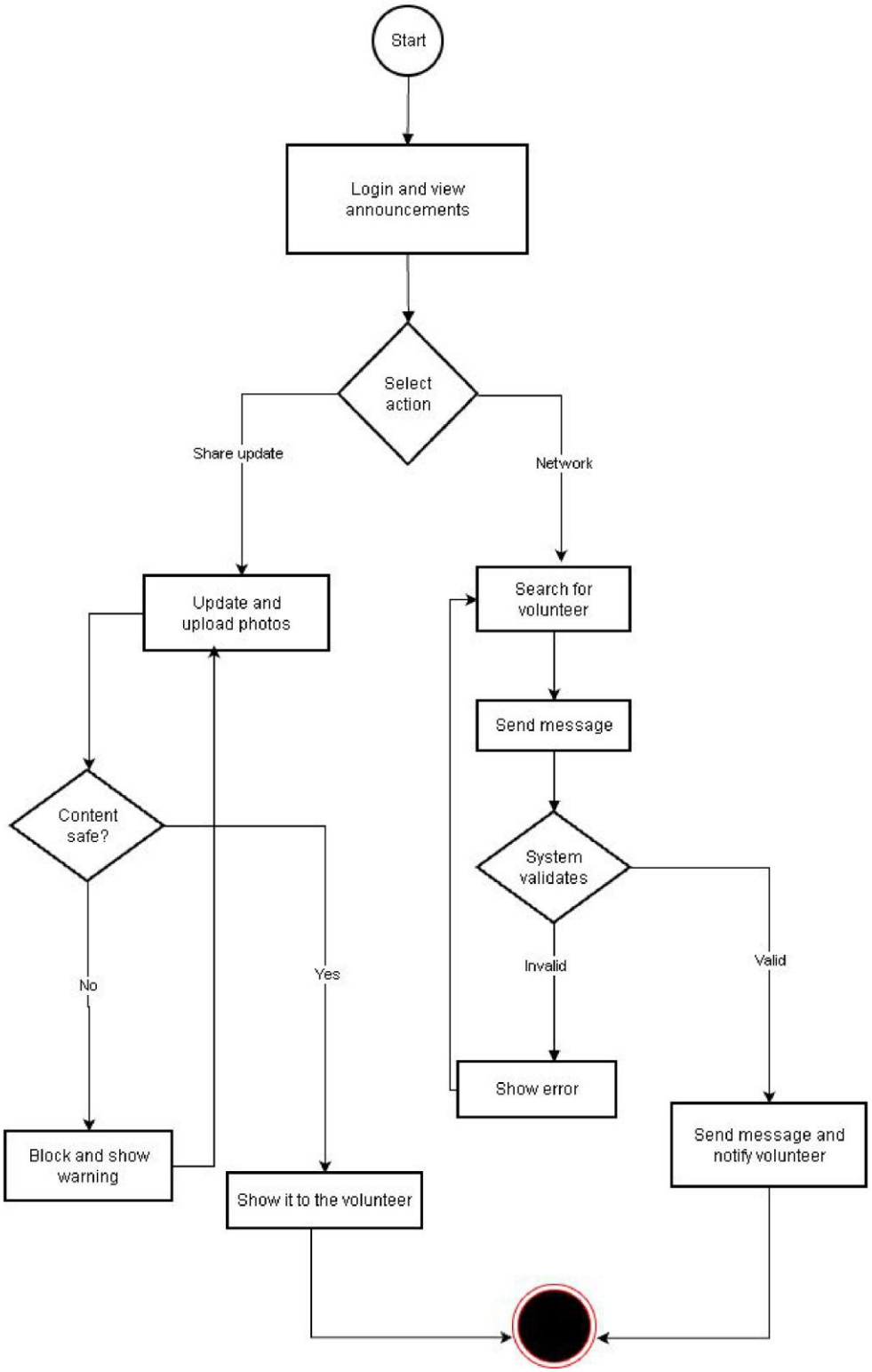
6.2 ACTIVITY DIAGRAMS:

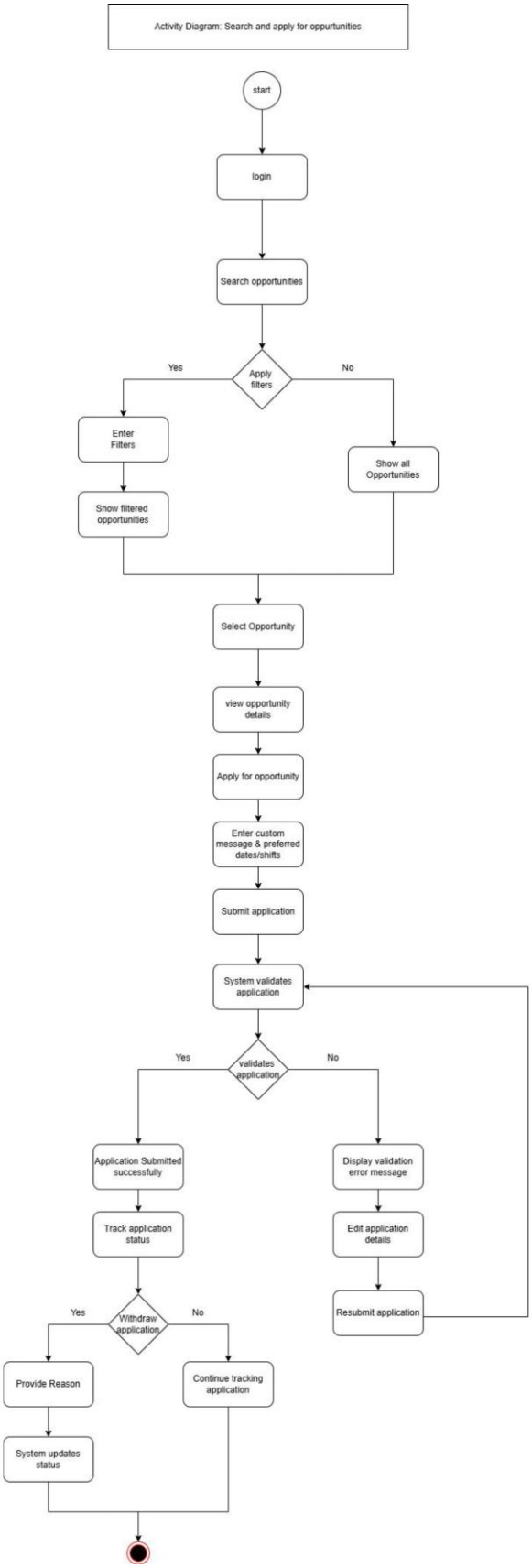

Volunteer Management
View Volunteer list and available programs
Verify Volunteer Credendtials and conduct background checks
Credentials Verified?
NO
YES
Reject Volunteer
Approve Volunteer for Program
set travel radius and transportation preference
Match Volunteers to Opportunities based on skills
Suitable match found?
YES
NO
Assign volunteers and allocate resources to programs
Search for alternative volunteers
Send assignment notifications to volunteers
Volunteer accepts assignment
Track volunteer hours
Hours Logged Automatically
YES
NO
System records check in and check-out times
Volunteer submits hours manually
Admin reviews and approves hours
Update volunteer records
Generats activity reports


Activity diagram : Manage Users
Start
Admin login
Select Manage Users
Create User
Enter User Details
Details Vaild?
Yes
No
Create account
Display Error
Show Confirmation
Edit User
Select Existing User
Modify User Details
Update User
Show Confirmation
Deactivate User
Confirm?
No
Yes
Deactivate Account
Cancel Operations
Assign Roles & Permissions
Select User
Assign Role & Set Permissions
Save Changes
Reset Password/ Manage Security
Select User
Reset Password / Update Security Settings
Notify User
Log Action

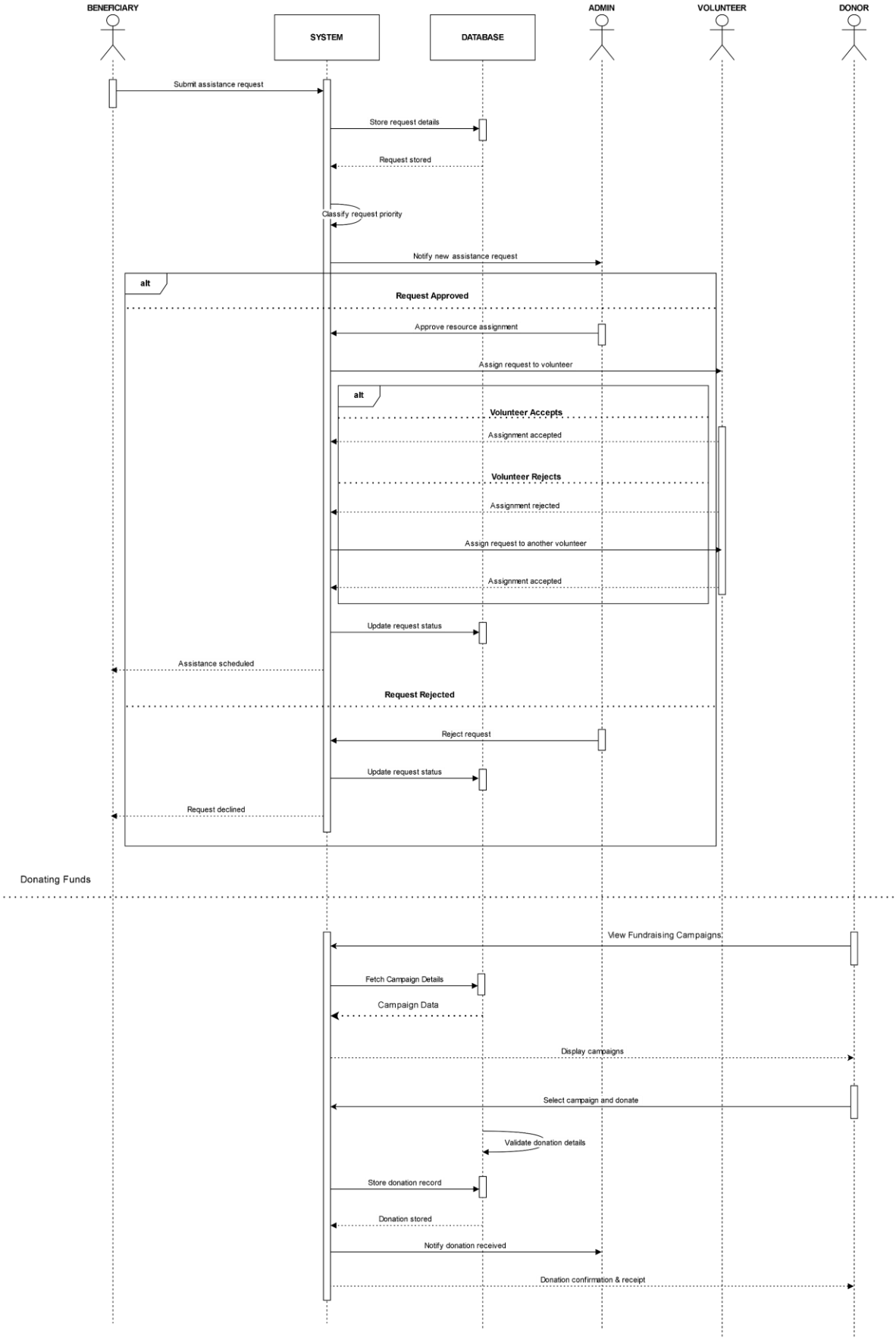

Register and apply for assistance
Start
Login
Complete Registration and Enter Household Information
Provide Demographic Info for program Eligibility Assessment
System Assessing Program Eligibility
Is Household Eligible?
No
Yes
System Checks for Existing Applications
Application Exists?
Yes
No
Specify Accessibility Needs
Update Emergency Contact Information
Manage Privacy Settings and Provide Data Consent for data sharing
Consent Provided?
No
Yes
Display Warning: Consent Required
System confirms Complete Application
Is Data Valid?
No
Yes
Display Error Message
Generate Tracking ID
Edit Details
Data is Stored
Send Confirmation to Applicant
Show Existing Application Status
Display Ineligibility Notice and show Reasons

Volunteer communication and network




Activity Diagram: Search and apply for oppurtunities
start
login
Search opportunities
Apply filters
Yes
No
Enter Filters
Show filtered opportunities
Show all Opportunities
Select Opportunity
view opportunity details
Apply for opportunity
Enter custom message & preferred dates/shifts
Submit application
System validates application
validates application
Yes
No
Application Submitted successfully
Track application status
Withdraw application
Yes
No
Provide Reason
System updates status
Continue tracking application
Display validation error message
Edit application details
Resubmit application

6.3 SEQUENCE DIAGRAMS:



7. APPENDICES

This section provides supplementary information related to the Volunteer and Resource Coordination System that supports the overall functionality but does not form part of the core system requirements.

Appendix A: Hardware Requirements

▪ A device such as a computer, laptop, tablet, or smartphone with internet connectivity.
▪ A reliable server with sufficient processing power, memory, and storage to handle user data.
▪ Secure backup hardware or cloud storage to prevent data loss.

Appendix B: Software Requirements

▪ A modern web browser (Chrome, Edge, Firefox, etc.) for accessing the system.
▪ Database management system to store volunteer, donor, and beneficiary information.
▪ Secure authentication system to protect user accounts and sensitive data.
▪ Basic operating system support (Windows, Linux, or macOS) for deployment and maintenance.

8. FUTURE SCOPE:

▪ Add automated notifications for volunteer assignments, resource updates, and beneficiary requests.
▪ Develop a mobile application to improve accessibility for volunteers, donors, and administrators.
▪ Implement intelligent volunteer matching based on skills,

availability, and location.

▪ Integrate secure online donation and real-time tracking features for better transparency.

9. CONCLUSION:

The Volunteer and Resource Coordination System helps NGOs manage volunteers, resources, donations, and beneficiary requests in an organized way. It reduces manual work, saves time, and improves communication between all users.

By using this system, NGOs can allocate resources properly and ensure that help reaches the right people at the right time. Overall, the system provides a simple and effective solution for better NGO management and supports their mission to serve society.